

LESSON 4.11

INTERPRETING AND COMPARING ABSOLUTE VALUES IN THE REAL WORLD



LESSON INTRODUCTION

MA.6.NSO.1.4 Solve mathematical and real-world problems involving absolute value, including the comparison of absolute value.

LEARNING TARGETS

- I can compare the absolute value of two numbers in a real-world context.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.7.NSO.2.1 Solve mathematical problems using multistep order of operations with rational numbers, including grouping symbols, whole-number exponents, and absolute value.

ESSENTIAL QUESTIONS

- How do you compare absolute values in real-world contexts?
- How do you use inequality symbols to compare absolute values?

LESSON NARRATIVE

Students compare absolute values in real-world contexts. Given written descriptions of real-world situations, students interpret and compare absolute values.

COMPONENT SUMMARY

4.11.1 WARM-UP (~5 MIN.)

Interpret and compare elevation absolute values

4.11.2 GUIDED INSTRUCTION (10-15 MIN.)

Learn how to interpret and compare absolute values in context

4.11.3 COLLABORATION (10-15 MIN.)

Collaboratively interpret and compare elevation absolute values

4.11.4 YOUR TURN! (10 MIN.)

Practice interpreting and comparing absolute values in contexts

4.11.5 WRAP-UP (~5 MIN.)

Demonstrate understanding of lesson learning goals

RESOURCES

GLOSSARY

Key Terms:

- Profit

Additional Terms:

- N/A

MATERIALS

- (Optional) Pictures or video clips of real-world contexts
- (Optional) Chart paper & markers

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle with interpreting the meanings of the words elevation, altitude, and sea level.
- Students may struggle with interpreting the meanings of the words revenue, expenses, and profit.

THINGS TO CONSIDER

- Consider connecting the terms “gross income,” “expenses,” and “net income” for an individual to the terms “revenue,” “expenses,” and “profit” for a business.

LITERACY AND LANGUAGE ROUTINES

Three Reads

In 4.11.3 Collaboration, students analyze a written description of elevations. The routine enhances student comprehension skills. Use to ensure students accurately interpret and compare the absolute values described in the written description.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

Students work with a partner in 4.11.3 Collaboration to interpret and compare absolute values related to elevation. Ask students to justify their answers and explain their strategies to their partner. This discussion will help students reflect on the efficiency of their strategies and accuracy of their solutions.

DIFFERENTIATE INSTRUCTION

Provide sketching paper or gridlines (mini coordinate planes) for students to utilize number lines during 4.11.3 Collaboration and 4.11.4 Your Turn! activities. You will be modeling the use of a number line during 4.11.2 Guided Instruction. While students do not need to produce a number line, this can often be a helpful strategy to provide students with a different entry point to make meaning of the context.

4.11.1 WARM-UP: ABSOLUTE ELEVATIONS

Component Overview: Students use the inequality symbols to compare absolute values. Then students explain the meaning of the inequalities in terms of the elevations of Death Valley and Tallahassee.

DIFFERENTIATE INSTRUCTION

Consider showing images or video clips of Death Valley, California, and Tallahassee, Florida, to help students understand the elevations of the locations.

Consider providing sentence stems or answer choices for question 3.



4.11.1 Warm-Up: Absolute Elevations

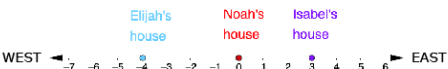
Death Valley, a national park in California, is at an elevation of -282 feet. The city of Tallahassee, Florida is at an elevation of 203 feet.

1. Use an inequality symbol, $<$ or $>$, to compare these two elevations.
 $-282 < 203$ or $203 > -282$
2. Use an inequality symbol, $<$ or $>$, to compare the absolute values of these two elevations.
 $|-282| > |203|$ or $|203| < |-282|$
3. Explain what these two inequality statements mean in terms of the elevations of Death Valley and Tallahassee.
Answers will vary. Based upon elevation, Tallahassee has a higher elevation than Death Valley, but Death Valley is farther away from sea level than Tallahassee.



4.11.2 Guided Instruction: Comparing and Interpreting Absolute Value in Context

- Recall this scenario from a previous lesson with some additional information. Elijah, Noah, and Isabel live on the same street that runs west to east. The locations of their houses are modeled on the number line, where each unit represents 1 kilometer.



- Noah's house is located at 0 on the number line model. Represent the distance from Noah's house to both Isabel's house and Elijah's house using absolute value.

	Elijah's House	Isabel's House
Distance from Noah's House	$ -4 $	$ 3 $

- How do the absolute values in the table compare to each other?

$|4| > |3|$

- Isabel lives 3 kilometers to the

east

west

 of Noah. Elijah lives 4 kilometers to the

east

west

 of Noah. Isabel lives

closer to

farther from

 Noah than Elijah does.

4.11.2 GUIDED INSTRUCTION: COMPARING AND INTERPRETING ABSOLUTE VALUE IN CONTEXT

Component Overview: Students learn how to interpret and compare absolute values in contexts. Students use a number line to aid in understanding their comparisons. The contexts explored include locations east and west of a house and the revenue, expenses, and profit of a lawn mowing business.

TEACHER MOVES

During instruction, note the advantages and disadvantages of using horizontal and vertical number lines in different contexts.

ENRICHMENT

- What does zero represent in the context?
- What represents positive values?
- What represents negative values?

4.11.2 GUIDED INSTRUCTION: COMPARING AND INTERPRETING ABSOLUTE VALUE IN CONTEXT (CONTINUED)

DIFFERENTIATE INSTRUCTION

Consider having students create reference cards with vocabulary terms from the contexts explored in this lesson. Include definitions, examples, and visual models to improve student comprehension.

2. Landon starts a lawn-mowing business to earn money. All of his profits are being saved for college.

Profit is the difference between the amount earned (revenue) and the amount spent.

His father lends him a lawnmower to use for free, but he must pay for the gas and other supplies. For each lawn, his expenses are \$7.30, and he earns an average of \$40.00.

- a. In this situation, revenue is represented with a ☐ positive ☐ negative value and an expense is represented by a ☐ positive ☐ negative value.

- b. Write a value to represent each situation.

Landon's Finances	Value
Expenses per lawn	-\$7.30
Revenue per lawn	\$40.00

- c. Use an inequality symbol, < or >, to compare the absolute values of Landon's expenses to revenue for each lawn he mows.

$$|\$40.00| > |-\$7.30|$$

- d. How does the inequality show that Landon will make a profit?

Answers will vary. The revenue he brings in is greater than his expenses, so he will make a profit.

- e. To make a profit, total expenses of a business must be ☐ greater than ☐ less than total revenue. A business is not profitable when the total ☐ expenses are ☐ revenue is greater than the ☐ expenses. ☐ revenue.



4.11.3 Collaboration: Exploring Additional Context

1. An underwater cave has an elevation of -118 feet. Here are the descriptions of four people's elevations in relation to the cave.
- Clare's elevation is a positive value. Clare is farther from sea level than the cave.
 - The value of Andre's elevation is smaller than the value of elevation of the cave. Andre is a farther distance from sea level than the cave.
 - Kiara's elevation is greater than the elevation of the cave. Kiara is closer to sea level than the cave.
 - Lin's elevation is the same distance away from sea level as the cave.
- Work with your partner to complete the following table.
- Write a possible elevation for each person based on the description.
 - Use $<$, $>$, or $=$ to compare the elevation of each person to the elevation of the cave.
 - Use absolute value to compare the distance each person is from sea level to the distance the cave is from sea level.

The first row has been completed for Clare.
Answers will vary. Possible answers below.

	Possible Elevation	Comparing Elevations	Comparison
Clare	153 feet	$153 > -118$	$ 153 > -118 $
Andre	-119 feet	$-119 < -118$	$ -119 > -118 $
Kiara	-52 feet	$-52 > -118$	$ -52 < -118 $
Lin	118 feet	$118 > -118$	$ 118 = -118 $

2. Priya says her elevation is less than the cave's and she is closer to sea level. Is this possible? Explain your reasoning or draw a picture.
Answers will vary. No. If the elevation is less than the cave then its distance will be farther away from sea level.

4.11.3 COLLABORATION: EXPLORING ADDITIONAL CONTEXTS

Component Overview: Students collaboratively interpret and compare elevation absolute values. Ask students to justify their answers and explain their strategies to their partner. This discussion will help students reflect on the efficiency of their strategies and accuracy of their solutions.

TEACHER MOVES

Three Reads

Students analyze a written description of elevations. The routine enhances student comprehension skills. Use this routine to ensure students accurately interpret and compare the absolute values described in the written description.

DIFFERENTIATE INSTRUCTION

Remind students to highlight or underline key information from the written description of the elevations. If students struggle to compare the absolute values, prompt them to create a number line.

4.11.4 YOUR TURN!

Component Overview: Students practice interpreting and comparing absolute values in contexts. The contexts explored include finance and elevation.

DIFFERENTIATE INSTRUCTION

Prompt students to sketch number lines to represent the situation and make meaning in context, if needed.

QUESTIONING

If students finish early, prompt them to write real-world problems involving comparing absolute values. Ask students to switch problems and compare the absolute values in each other's problems.



4.11.4 Your Turn!

1. A bank transaction occurs when money is passed between a person and the bank. Liliana has recently made two transactions at her bank. On Monday, she deposited \$127 into her bank account. A few days later, she withdrew \$140 from her account.
 - a. Write a value that could be used to represent each value in this situation.

Liliana's Bank Transaction	Value
Deposit	\$127
Withdrawal	-\$140

- b. Use an inequality symbol, $<$ or $>$, to compare the absolute values of Liliana's transactions.
 $|\$127| < |-\$140|$ OR $|-\$140| > |\$127|$
 - c. What does this inequality statement indicate about the amount of money in Liliana's bank account after she made both transactions?
Answers will vary. She deposited less than she withdrew from the account. This result is an overall decrease in her account balance after both transactions.
2. An octopus is 47 feet below sea level at the same time a dolphin is at 108 feet below sea level.
 - a. Write a value to describe the position of each sea animal relative to sea level.

Octopus	Dolphin
-47 feet	-108 feet

- b. Use an inequality symbol, $<$ or $>$, to write an inequality comparing the distance each sea animal is from sea level.
 $|-47| < |-108|$ OR $|-108| > |-47|$
 - c. What does this inequality statement indicate about the locations of each sea animal?
Answers will vary. The dolphin is farther from the surface of the water than the octopus.



4.11.5 Wrap-Up: Ticket Out the Door

- The table shows the elevation, in meters (m), of five locations.

Location	Elevation (m)
Vpadina Kaundy, Kazakhstan	-132
Laguna del Carbón, Argentina	-105
Zuidplaspolder, Netherlands	-7
Fort Myers, Florida	2
Britton Hill, Lakewood, Florida	105

For each pair, use $<$, $>$, or $=$ to compare their distances from sea level.

Zuidplaspolder, Netherlands	$ -7 > 2 $
Fort Myers, Florida	

Britton Hill, Lakewood, Florida	$ 105 = -105 $
Laguna del Carbón, Argentina	

Laguna del Carbón, Argentina	$ -105 < -132 $
Vpadina Kaundy, Kazakhstan	

4.11.5 WRAP-UP: TICKET OUT THE DOOR

Component Overview: Students demonstrate their mastery of the lesson learning targets. Use data from this formative assessment to determine which students need remediation before the next lesson.

QUESTIONING

- Consider a project-based learning experience for students as homework, extending the 4.11.5 Wrap-Up.
- Map elements of your bedroom, with the level of your pillow or the door handle as “sea level” in the context of elevation.
 - Provide the “sea level” of at least five objects in your room.
 - Students can use different measuring tools as long as a key is provided.